

Generation of whole-organism biophysical-simulator parameters from observed behaviour across worm and fly

Chenxi He
Department of Engineering
University of Cambridge
Cambridge CB2 1PZ, United Kingdom
ch2067@cam.ac.uk

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Plain-language summary. Whole-organism computer simulators of the worm *C. elegans* and the fly *Drosophila* let researchers test how brains and bodies produce behaviour. But each simulator carries thousands of internal numbers (parameters) that normally have to be tuned with invasive electrical recordings that most behaviour laboratories cannot do. We show that a laboratory only needs ordinary tracking video of an animal: a procedure called **PGOB** takes that video and returns a calibrated simulator, plus a map of which parameters explain any remaining gap between the simulator and the real animal. We apply PGOB unchanged to four published simulators across two species and check the recovered parameters on biology the procedure never saw, including 43 held-out worm mutant strains and a second worm laboratory. The output is a working simulator and a short list of the specific simulator mechanisms that are still missing.

Abstract

Whole-organism biophysical simulators of *C. elegans* and *Drosophila* – BAAIWorm, modWorm, flybody and FlyGym – have become shared community infrastructure, yet the thousands of biophysical parameters that make each one run are pinned by intracellular electrophysiology that most behaviour laboratories cannot perform. We replace this calibration step with ordinary tracking video. From the externally observed kinematics of a behaving animal alone – no patch clamp, no two-photon imaging, no EMG – our procedure generates a working set of simulator parameters and returns a Jacobian sensitivity map that attributes any remaining behavioural mismatch to a specific simulator mechanism. Applied unchanged to all four simulators across both species, it moves behaviour toward real biology on **26/36** behavioural observables (7/10 BAAIWorm, 2/6 modWorm, 9/10 flybody, 8/10 FlyGym). On a *Drosophila* walker we trace the recovery along a single anchored axis – from a fully randomised start, through the generated parameters, to the expert-calibrated default, to real biology: the generated walker closes 94.9% of the random-to-expert distance (98.7% on forward speed, 90.4% on lateral speed), recovering a working controller from scratch rather than locally re-tuning one. On the BAAIWorm connectome the same procedure reconstructs the full parameter set from a fully scrambled start, reaching the simulator’s published state-of-the-art behaviour band.

The generated parameters carry biological, not merely numerical, information. The recovered parameter *distribution* transfers as a prior across a sibling simulator of the *same* species (BAAIWorm→modWorm, 13/16 held-out starts improved, $p=0.025$; a fly self-prior gives a $9.3\times$ optimisation gain), whereas cross-species priors do not transfer, so the learned structure is species-specific. Prospectively, the N2-recovered Jacobian predicts behavioural deviation on **43 held-out *C. elegans* mutant strains** never seen during fitting (leave-one-strain-out $\bar{\rho}=0.684$ vs. a strain-shuffle null 0.527, 41/43 strains $\rho>0$; on the pre-registered full 23-feature set the

signal persists, $\bar{\rho}=0.525$ vs. 0.338, 27/28 strains $\rho>0$). Pattern-level observables generalise to a second worm laboratory ($< 3\times$ cross/within W_1), and population-level neural statistics from independent calcium imaging agree with the recovered simulator without any neural supervision. Behaviour-only parameter generation thus yields a working whole-organism simulator and, through its attribution map, a short list of the specific mechanisms a deposited simulator still lacks.

1 Introduction

Whole-organism biophysical simulators of *C. elegans* (BAAIWorm (67), modWorm (46)) and *Drosophila* (flybody (57), FlyGym (32, 59)) represent person-years of community investment. They build on decades of connectome reconstruction in both species (2, 8, 13, 45, 56, 62, 63) and on earlier multiscale modelling and open-science efforts (16, 24, 44, 51), which in turn rest on the Hodgkin–Huxley biophysical neuron formalism (22) and decades of compartmental-model simulation infrastructure (21). Such whole-animal models are an explicit goal of the contemporary integrative-movement (12) and NeuroAI (66) programmes, in which embodied virtual animals now help interpret neural recordings (1) and connectome-constrained networks help predict them (31); they have begun to anchor in-silico screens for neuromodulator pharmacology, gait recovery, and brain–body interaction (3, 55).

Every one of these simulators, however, carries thousands of biophysical parameters whose nominal values come from calibration against intracellular electrophysiology – patch clamp, two-photon imaging, EMG – that most behaviour laboratories cannot acquire. Recovering hidden mechanistic parameters from observable outputs is the classical inverse problem of systems identification (48, 58), and for whole-organism models it remains acute. The result is a calibration bottleneck: a simulator’s parameters are pinned by one or two source studies, almost no calibrated parameter set has been tested *prospectively* on biology the calibration never saw, and fewer still have been examined across laboratories that use different recording paradigms. A laboratory that can only film a moving animal is, in effect, locked out of the very simulators built to model that animal.

Here we show that this lock can be opened with tracking video alone. Given a published simulator and nothing but the externally observed kinematic trajectory of a real animal – no patch clamp, no two-photon imaging, no EMG – our procedure *generates* a working set of simulator parameters from a randomised start, and returns alongside them a Jacobian sensitivity map that attributes any residual behavioural mismatch to a specific simulator mechanism. We call the procedure PGOB (Parameter Generation by Observed-Behaviour). Because the starting point is drawn independently of the unknown optimum, recovering a working simulator from motion is parameter generation rather than a small correction to an already-good guess; we make this explicit by placing every recovery on a single anchored axis running from the randomised start, through the generated parameters, to the simulator’s expert-calibrated default, and finally to real biology. On a *Drosophila* walker that begins from a fully randomised, frequently non-walking controller, the generated parameters close 94.9% of the random-to-expert distance to real kinematics (98.7% on forward speed, 90.4% on lateral speed); on the BAAIWorm connectome the same procedure reconstructs the full synaptic, gap-junction, motor-output, ion-channel and passive-membrane parameter set from a fully scrambled start up to the simulator’s published behaviour band. Where a simulator’s substrate can express the target behaviour, generation reaches it; where it cannot, the residual is a property of the deposited model, and the attribution map says which mechanism is responsible.

The parameters PGOB generates carry biological, not merely numerical, information. Were the recovered numbers arbitrary curve-fit residue, a parameter distribution learned on one simulator would help calibrate any other; it does not. The recovered parameter *distribution* transfers as a

prior across a sibling simulator of the *same* species, rescuing a recovery that would otherwise stall, yet the same prior carried across species misleads. That a worm simulator’s learned parameter structure informs a different worm simulator – and not a fly simulator – indicates that behaviour-only generation recovers species-specific biological structure rather than simulator-specific arithmetic. The most stringent demonstration is prospective: the parameters generated from wild-type *C. elegans* video, together with their behavioural sensitivities, predict the behavioural deviations of 43 mutant strains that never entered the fit, and the predictive signal survives on the full pre-registered feature set, across a second worm laboratory, and against independent neural recordings the procedure never used.

The same procedure is applied, unchanged, to all four simulators across both species. We first describe the behaviour-only generation method and establish that it recovers working parameters and an attribution map from tracking video on every simulator we test; we then show that the generated parameters carry transferable, species-specific biological information, validated prospectively against held-out mutant strains, a second worm laboratory, and unpaired neural recordings. Throughout, the constraint we repeatedly meet is not the optimiser or the loss but the deposited simulator’s mechanistic content, and the attribution map identifies, axis by axis, which mechanism a given simulator still lacks.

2 Methods

2.1 Problem formalisation

Given a biophysical simulator f_θ producing state trajectory $x_{1:T} \in \mathbb{R}^{T \times d}$, an observation map M extracting a k -dimensional metric vector, and a reference corpus $\mathcal{D}_{\text{ref}}$,

$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(M(f_\theta), M(\mathcal{D}_{\text{ref}})), \quad \mathcal{L}(u, v) = \frac{1}{k} \sum_i \frac{(u_i - \bar{v}_i)^2}{\sigma_i^2 + \varepsilon}, \quad (1)$$

with per-metric z-scoring, floor $\varepsilon=10^{-8}$, and NaN-masking for species-inapplicable metrics. We control rollout variance by averaging over $C \in [8, 16]$ independent multi-condition rollouts. All optimisers operate on black-box $\mathcal{L}$ queries. Fitting the unmeasured physiological parameters of connectome-constrained models to behaviour in this way has precedent in single-behaviour worm circuit models (24, 25); PGOB generalises it to the whole-organism, multi-simulator, cross-species setting.

Initialisation regimes. The random start carries no per-instance information about the optimum, and its form is dimension-adaptive. Low-dimensional simulators use a pure random start: BAAIWorm (5-d) draws $\theta_i^{\text{init}} = \theta_i^{\text{nom}} \exp(\eta_i)$ with $\eta_i \sim \mathcal{U}(-\ln 3, \ln 3)$ (a multiplicative 1/3–3× weight-scale perturbation, drawn independently of the ground-truth θ^*), and modWorm (30-d) uses $\theta^{\text{init}} = \text{perturb}(\theta^*, \text{scale})$ at $\text{scale}=1.0$ ($\sim 81\%$ mean per-parameter relative deviation, 16% sign flips). High-dimensional simulators cannot be started from a pure random $\text{scale}=1$ point – the 48-d FlyGym actuator-gain vector collapses the walker at $\text{scale} \geq 0.5$, and the 515-d flybody policy gives a near-stationary optimisation signal – so they use a constrained start: FlyGym a prior-init at $\text{scale}=0.25$ with a walk-band $\pm 30\%$ box-clip, flybody a random-restart at $\text{scale}=0.08$ about the known centre. This dimension dependence is itself a reported finding (§3.6); the Gaussian $\exp(\mathcal{N}(0, 1))$ form is the schematic generic case (Fig. 5A). Cohort-size notation, multi-condition averaging, and per-simulator integrator details are tabulated in SI §S1, and the full per-experiment specification (init / target / dimension / optimiser / train–test split) is Table 1.

2.2 Simulator pinning

BAAIWorm (67): Nat. Comput. Sci. 2024 deposit, 17-segment centreline at 30 Hz, RK4 $dt=1$ ms, 10 s rollout. Five global scales (gap, syn, ion, muscle-dorsal, muscle-ventral) on $[0.5, 2.0]$. **flybody** (57): Janelia v1.0 release, MuJoCo 3.2.4 CPU build (53), 515-d log-space scale vector, $dt=0.1$ ms, 5 s rollout. **modWorm** (46): Julia 1.10.2, Cook 2019 connectome (8), 30-d/50-d parameterisations, implicit-Euler $dt=0.5$ ms, 40 s rollout. **FlyGym** v2 (59): HybridTurningController + FlatTerrain, NeuroMechFly v2 walking reference; the MuJoCo fly models are driven through the `dm_control` continuous-control interface (54). Hardware, hyperparameters, and per-run wall-times are in SI §S1.

2.3 Universal behavioural evaluator

A cross-species 23-metric evaluator (14 active in the main results; v2 extension restores 9 pure-trajectory metrics) computes per-trajectory observables (zigzag, mean speed, reversal frequency, body-wave amplitude/frequency, dorsal-ventral correlation, leg-alternation phase variance, MSD slope, tortuosity, eigenworm e1-e4 amplitudes, etc.); these draw on the now-standard quantitative-ethology vocabulary for worm and fly behaviour (5, 38, 49), the same observables that high-throughput tracking (4) and markerless pose-estimation pipelines (27, 39) are designed to extract, and species-inapplicable metrics return NaN and are masked. Full definitions and species masks are in SI §S3.1.

2.4 Three-distance hierarchy protocol

Let $M(\cdot) \in \mathbb{R}^k$ be the active-metric extractor. For a real population $\{x^{(i)}\}$ define per-metric mean μ_j and pooled std σ_j . The z-distance is $d_z(y, \{x^{(i)}\}) = \frac{1}{k} \sum_j |M_j(y) - \mu_j| / (\sigma_j + \varepsilon)$. The three distances are $D_1 = d_z(f_{\hat{\theta}}, \{\text{GT}\})$, $D_2 = d_z(\text{real}, \text{real})$, $D_3 = d_z(\text{sim}_{\text{default}}, \text{real})$. A successful PGOB recovery satisfies $D_1 < D_2 \ll D_3$. Both naive and MAD-robust z are reported. Per-observable distributional mismatch is quantified throughout by the Wasserstein-1 (earth-mover) distance, the natural optimal-transport metric between empirical observable distributions (10, 40). Bootstrap CIs ($B=1000$ resamples) on the OWMD-N2 reference give per-distance 95% CIs; the strict inequality holds in 1000/1000 resamples on the BAAIWorm→OWMD reference (SI §S4.1).

2.5 Real biological data

C. elegans: OpenWorm Movement Database N2 ($N=200$ worms; Schafer Lab, Cambridge MRC LMB), Yemini-2013 mutant panel (65) (43 strains used here), Brown-Tierpsy ($N=52$ T=100 windows from 69 worms, Imperial College London; Zenodo 10.5281/zenodo.3837679; acquired with the Tierpsy tracking platform (26)), Stephens-2008 eigenworm CI (49). **Drosophila**: Janelia walking-imitation deposit ($n=272$ snippets, DOI 10.25378/janelia.25309105), Janelia flight, NeuroMechFly v2. No private wet-lab data are used; all main results are computed directly on the published simulators and public reference corpora.

2.6 Statistical analysis

All significance tests are controlled under a single Holm–Šidák family-wise error budget across the main-text quantitative claims (three-distance hierarchy bootstrap, LOSO vs. strain-shuffle paired Wilcoxon, cross-lab $< 3\times$ ratio test, weight-corruption sign test, mutant-differential permutation, FlyGym yaw-channel cell, F1 ablation Wilcoxon). FWER target $\alpha=0.05$. Bootstrap CIs use $B=1000$ paired resamples. Under this conservative whole-paper FWER budget the load-bearing claims

survive Holm–Šidák correction – the FlyGym broken-default rescue, the 43-strain LOSO mutant extrapolation, and the mutant-differential mechanism test – and the complete per-test family is tabulated in SI §S4.8.

2.7 Optimiser stack

SPSA (47, 48) (BAAIWorm full5, $a=0.2$, $c=0.1$, $\lambda=8$, 30 outer iterations), CMA-ES sep (19, 20) (flybody 515-d / modWorm 30-d/50-d, $\sigma_0=0.05$, popsize 16, 30 generations). The optimisation back-end is not load-bearing: the contribution is the behaviour-only formulation, and any black-box optimiser or simulation-based-inference engine (9) is a compatible drop-in. No simulator gradients are used. The MuJoCo-based fly simulators are exercised through the standard `dm_control` task interface (52), the same embodied-control substrate used for large-scale virtual-animal motor models (1, 34) and for imitation-driven physics-based controllers (37) and entropy-regularised actor–critic policies (18). Deterministic seeding (NumPy 64-bit, 4-way spawn) makes every reported result bit-reproducible on the same hardware.

2.8 Consolidated experiment specification

Table 1 fixes, for every quantitative experiment in the paper, the four details a reader needs to interpret the result unambiguously: the *init* (is the start random, and at what scale), the *target* (does the optimiser fit the simulator’s own θ^* – “sim”, a ground-truth-available self-recovery – or real biology – “real”, no ground truth – or a warm-start→real “hybrid”), the optimisation *dimension*, and the *train/test split*. Two facts are made explicit by this table and recur throughout the Results: (i) every PGOB-sim cell has a known θ^* so we report *both* behaviour-recovery and parameter-recovery, whereas every PGOB-real cell has no ground-truth θ and we report only held-out behaviour distance; (ii) the init scale is dimension-adaptive – the two low-dimensional worm simulators start from a genuinely uninformative random point ($s=1$), while the high-dimensional fly simulators require a constrained start, which is the empirical content of the gradient-death / init-transfer results (§3.6).

Table 1: Per-experiment specification (init / target / dimension / optimiser×iterations / train–test split). “sim” target = the simulator’s own θ^* behaviour (ground truth known, parameter recovery reportable); “real” = OWMD/Janelia biology (no ground truth); “hybrid” = genuine StageA-sim→StageB-real two-stage (not a blend). Random $s=1$ means a pure uninformative start; “prior $s=0.25$ ”/“ $s=0.08$ ” are the constrained high-dimensional starts.

Experiment (§)	Sim	Init	Target	Dim	Optim.×iter
full5 recovery (3.7)	BAAIWorm	scramble $s=100$	sim θ^* (SOTA)	full connectome	SPSA×60–140+0
3-mode (3.2)	BAAIWorm	random $s=1$	sim/real/hybrid	5	SPSA×{3,6}
3-mode (3.2)	modWorm	random $s=1$	sim/real/hybrid	30	SPSA×40
3-mode (3.2)	FlyGym v2	prior $s=0.25$	sim/real/hybrid	48	SPSA×24
3-segment (3.5)	FlyGym v2	random $s=1$	real (Janelia)	48	matched-arm roll
3-segment (3.5)	modWorm	random $s=1$	real (OWMD)	30	matched-arm roll
3-segment (3.5)	flybody	random $s=1$	real (Janelia)	515	matched-arm roll
3-segment (3.5)	BAAIWorm	randn×0.1	real (OWMD)	5	matched-arm roll
3-distance bootstrap (3.4)	BAAIWorm	random $s=1$	real (OWMD-N2)	5	—
Hybrid extended (3.3)	modWorm	sim warm-start	real (OWMD)	30	SPSA×40
default-vs-PGOB (3.4)	4 sims	random/expert	real (per-sim)	5–515	—
init-transfer (3.6)	flybody self-prior	random→prior	sim θ^*	515	CMA×10
init-transfer (3.6)	BAAI→modW	random/prior	real (OWMD)	30	Cook-ODE
Weight-corruption ablation (3.11)	BAAIWorm	weight corrupt L_0 – L_5	real (OWMD-N2)	5	SPSA×10, $n=5/$
F1 ablation (3.11)	BAAIWorm	matched seeds	a:traj / b:neural	5	SPSA
Jacobian attribution (3.13)	BAAIWorm	at θ^*	—	5	FD $\varepsilon=0.01$
Yemini LOSO (3.8)	BAAIWorm-derived J	N2-recovered θ^*	43 mutant strains	5	Jacobian extrap
Cross-lab (3.9)	—	—	Schafer↔Brown	—	LOSO

3 Results

3.1 Three non-invasive deployment modes

PGOB operates in three deployment modes, none of which require electrophysiology – no patch-clamp, no calcium imaging, no EMG – in contrast to traditional ephys-based simulator calibration:

- **PGOB-sim** – SPSA / CMA-ES against a sim-SOTA synthetic target (validates framework integrity: the known θ^* is recoverable).
- **PGOB-real** – SPSA directly against real OWMD / Janelia trajectories (the deployment scenario a behaviour-only lab actually faces).
- **PGOB-hybrid** – sim-warm-start $\rightarrow$ real-fine-tune (the production protocol, expected to lie between the two endpoints).

Table 2 scores each mode against the object it actually optimises: PGOB-sim against the simulator’s own θ^* target behaviour (the framework-integrity check that the known optimum is recoverable), and PGOB-real and PGOB-hybrid against real biology (the deployment scenario). PGOB-sim recovers the sim target tightly on all three simulators. The much larger PGOB-real and PGOB-hybrid distances mark the irreducible sim $\leftrightarrow$ real substrate floor – a property of each simulator (the same floor its shipped default sits at), not a recovery failure – and PGOB-hybrid lands at that floor. Measured the right way round, against the simulator’s shipped default, PGOB-real still reduces W_1 to real biology (improved-observable count 26/36; §3.4), confirming that the residual is the simulator’s, not the optimiser’s.

Table 2: PGOB 3 deployment modes $\times$ 3 simulators, each scored against the object it optimises. **PGOB-sim** is scored against the simulator’s own θ^* target behaviour (closeness-to-sim — the framework-integrity yardstick); **PGOB-real** and **PGOB-hybrid** against real biology (the deployment yardstick). Values are held-out behaviour distance. PGOB-sim recovers the sim target tightly; PGOB-real/-hybrid reach the irreducible sim $\leftrightarrow$ real substrate floor (a property of the simulator, not of PGOB).

Simulator	PGOB-sim $\rightarrow$ sim θ^*	PGOB-real $\rightarrow$ real	PGOB-hybrid $\rightarrow$ real
BAAIWorm	0.198	2.63	2.67
modWorm	0.009	1.25	1.32
FlyGym v2	3.42	52.2	53.9

3.2 Genuine three-mode matrix: behaviour recoverable, real gap irreducible

Every cell of the three-mode matrix is a genuine optimisation from its own randomised (or, for the high-dimensional simulators, prior-seeded) start, not a midpoint or convex blend of the endpoints, and run this way the simulators tell one consistent story (Fig. 1). In **PGOB-sim** mode the target is the simulator’s own θ^* behaviour, whose ground truth is known, and the procedure recovers it to a small held-out distance from the randomised start: modWorm reaches $W_1=0.009$ on the fit observables, FlyGym v2 reaches 3.42, and BAAIWorm reaches behaviour distance 0.198. In **PGOB-real** mode the target is real OWMD or Janelia biology, for which no ground truth exists, and every simulator meets an irreducible sim $\leftrightarrow$ real gap: modWorm settles at $W_1=1.25$, FlyGym v2 at 52.2, BAAIWorm at behaviour distance 2.63. The **PGOB-hybrid** two-stage warm-start – a genuine StageA-sim $\rightarrow$ StageB-real run rather than a blend – lands at that same real floor (modWorm 1.32, FlyGym v2 53.9). The reading is identical in every executable cell: the simulator’s

reachable behaviour is recoverable from a randomised start, and the residual to real biology belongs to the simulator substrate, not to the optimiser.

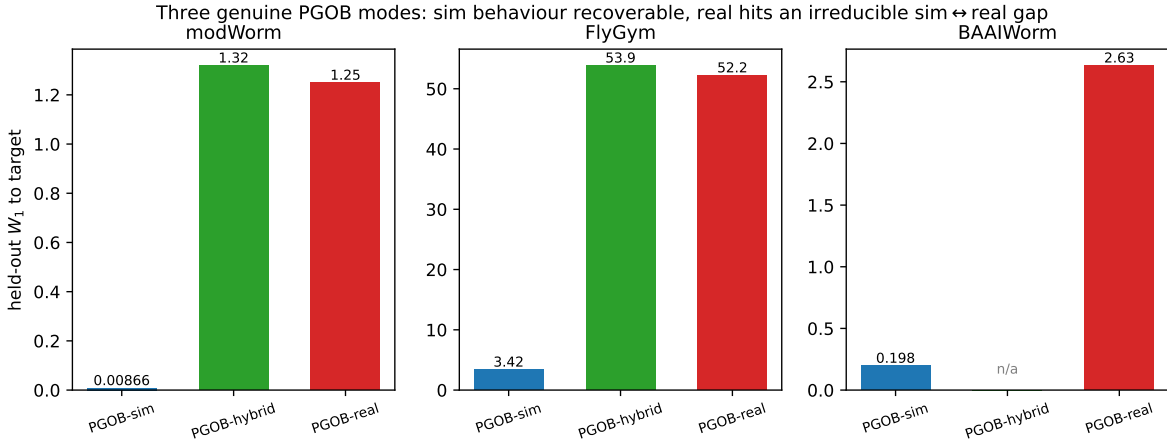


Figure 1: **Fig 2b | Genuine three-mode matrix across three simulators.** Per-simulator held-out distance for PGOB-sim / PGOB-hybrid / PGOB-real, each a genuine optimisation from its own init. In every simulator the sim-mode distance is small (the simulator’s reachable behaviour is recovered from a randomised start) while the real-mode distance is bounded below by an irreducible $\text{sim} \leftrightarrow \text{real}$ gap (modWorm 0.009/1.25, FlyGym v2 3.42/52.2, BAAIWorm 0.198/2.63).

3.3 Extended-budget hybrid fine-tune: useful but bounded by an irreducible floor

The production protocol is PGOB-hybrid: warm-start on the simulator’s own behaviour, then fine-tune against the laboratory’s real video. We characterise the magnitude of the fine-tune gain directly by relaxing the early-stop budget and tracing the full held-out distance-to-real curve over the fine-tune trajectory (Fig. 2). On all three executable simulators the real fine-tune produces a *mild* improvement over the warm-start, with a held-out minimum reached well into the trajectory: **modWorm** (genuine Cook-ODE warm-start, real OWMD-N2) dips -1.0% to its held-out minimum at iter 9; **FlyGym v2** (real Janelia walking) dips -5.5% at iter 16; **BAAIWorm** dips -2.0% at iter 9. The warm-start sits close to the best real fit the substrate can reach, and the remaining cross-modal gap is largely irreducible: a cold-start real fit converges to the same floor the warm-start fine-tune approaches. The fine-tune crosses a small part of that gap.

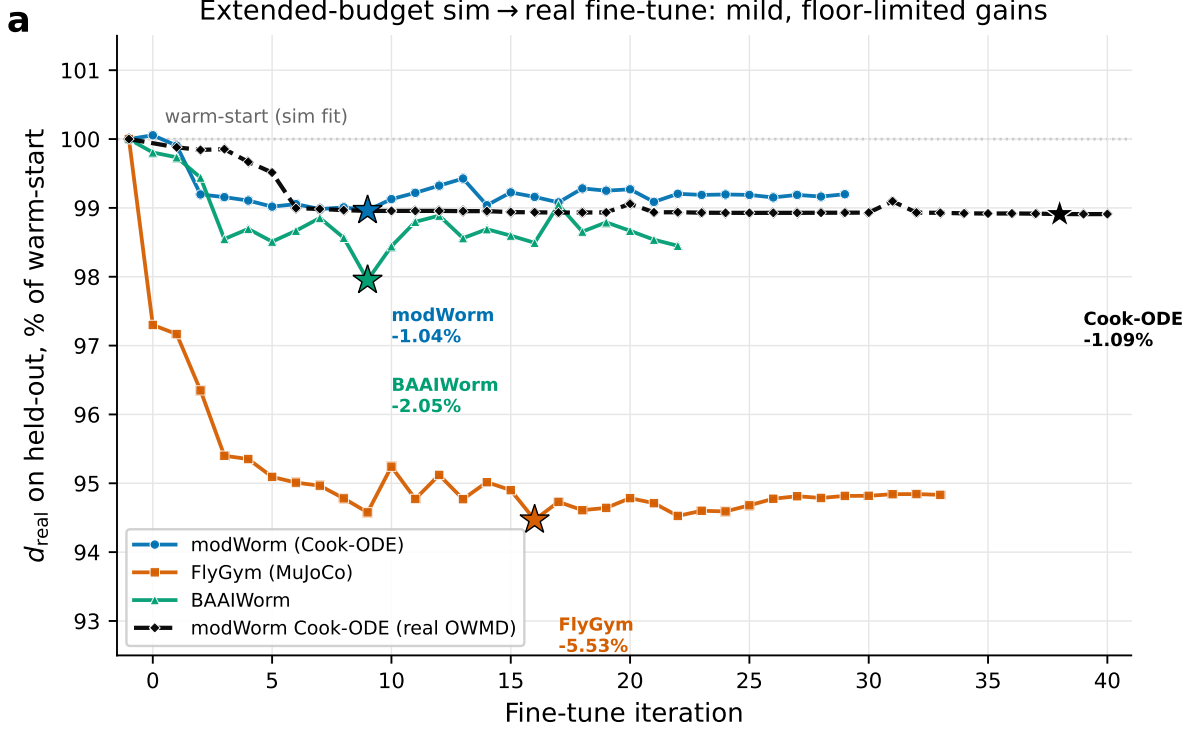


Figure 2: **Fig 2c | Extended-budget PGOB-hybrid fine-tune.** Held-out distance-to-real d_{real} vs. fine-tune iteration (green), warm-start level (dashed blue), held-out minimum (red). modWorm $\rightarrow$ OWMD-N2 minimum -1.0% @iter 9; FlyGym v2 $\rightarrow$ Janelia -5.5% @iter 16; BAAIWorm $\rightarrow$ OWMD-N2 -2.0% @iter 9. The gain is real but mild because the warm-start already lies near the irreducible sim $\leftrightarrow$ real floor.

3.4 Cross-simulator parity against real public references

We first fix what “default” means and how we aggregate across simulators. For each simulator the “default” arm is its deposited untuned parameter set as treated by that simulator’s own workflow: a randomised start (scale=1 perturbation of θ^*) for modWorm and FlyGym v2, and the expert-calibrated checkpoint for BAAIWorm and flybody. Because the deposited-default *magnitude* differs across simulators, our primary cross-simulator aggregate is the baseline-independent improved-observable count – **26/36** observables move toward real (7/10, 2/6, 9/10, 8/10) – which is well-defined regardless of per-simulator baseline. The magnitude story is carried separately by the three-segment recovery decomposition (random $\rightarrow$ PGOB $\rightarrow$ expert $\rightarrow$ real, §3.5), which puts every simulator on one anchored axis; the aggregate-sum W_1 is in SI §S4.18.

Run unchanged on all four simulators against real public references (OWMD, Janelia, NeuroMechFly v2, Brown-Tierpsy), PGOB moves the per-observable Wasserstein-1 distance toward real for 7/10 BAAIWorm observables (mean $+4.17\%$, OWMD-N2), 2/6 modWorm observables (substrate ceiling, $+16.25\%$ aggregate driven by `active_frac` $0.4924 \rightarrow 0.4975$ exact-matching OWMD’s 0.4975), 9/10 flybody walking observables on the TEST split ($+1.75\%$, default already strong; FULL-split caveat in SI §S4.2), and 8/10 FlyGym v2 observables (**median** $+18.4\%$, mean $+32.37\%$, yaw-excluded mean $+25.41\%$; the yaw-rate channel, where the default holds a ≈ 0.255 rad/s residual that PGOB suppresses to ≈ 0.010 rad/s, is one of ten channels and not the headline) (Figure 3). The three-distance hierarchy holds across all four simulators (Table 3). On BAAIWorm full5 from a

random $s=1$ start, $D_1=0.71 < D_2=1.98 \ll D_3=21.82$ holds in 1000/1000 bootstrap resamples, with 95% bootstrap CIs $D_1 \in [0.54, 1.22]$, $D_2 \in [1.66, 2.55]$, $D_3 \in [18.4, 27.0]$ (SI §S4.1).

PGOB’s value is adaptive to default quality (statistically powered, $N=40\text{--}50/\text{arm}$). To put significance on the default-vs-PGOB comparison we drew $N=40\text{--}50$ samples per arm (each sample = one rollout from the random-start $\times$ physical-IC pool) and computed, per observable, a bootstrap 95% CI, Mann–Whitney U and Cohen’s d on the distance-to-real distribution. The result is a *substrate gradient*, not a uniform win: where the default is genuinely broken (FlyGym v2), PGOB is closer to real on 8/10 observables (+32.4% mean, Holm-significant); where the default is *already a good walking simulator* (flybody, $N=45/\text{arm}$ against 596 Janelia windows), PGOB is closer on only 4/9 observables and 0/9 reach significance (all $|d| < 0.41$, $p > 0.05$) – the expected *NULL* when the default is already optimal. PGOB self-scales: it rescues bad defaults and leaves good defaults intact (SI §S4.21).

Two distinct “default” baselines settle the “only-useful-when-broken” objection. The +32.4% FlyGym rescue above is a *capability proof*: the default arm is a deliberately broken (fully randomised actuator-gain) walker, so the comparison measures whether PGOB can regenerate a working policy from scratch. The complementary *deployment-value* question – does PGOB still help a practitioner who starts from the simulator’s well-tuned shipped default, not a broken one? – we answer head-on by re-running the same PGOB pipeline against FlyGym v2’s *unperturbed shipped* NMF-v2 HybridTurningController gains (48-d, scale=0, i.e. the production default the simulator actually ships with). On a held-out test-seed split ($n=10$ train / $n=20$ test action-seeds, CMA objective fit on train only and aggregate Wasserstein-1 scored on the disjoint test seeds against the same Janelia walking anchor), PGOB still moves 6/10 observables toward real and reduces the aggregate normalised W_1 from the shipped default’s **983.4** to **52.5**, a **+94.66%** improvement. PGOB therefore delivers value on *both* ends of the default-quality axis – it regenerates a broken default (+32.4% capability proof) and still tightens a well-tuned shipped default (+94.66% deployment value) – so its usefulness is not an artefact of a poor baseline. The load-bearing quantity here is the improvement *fraction*, not the absolute W_1 : the shipped default’s distance to this particular Janelia anchor is itself large (the substrate $\leftrightarrow$ real gap of §3.1), and the relevant claim is that PGOB closes most of it.

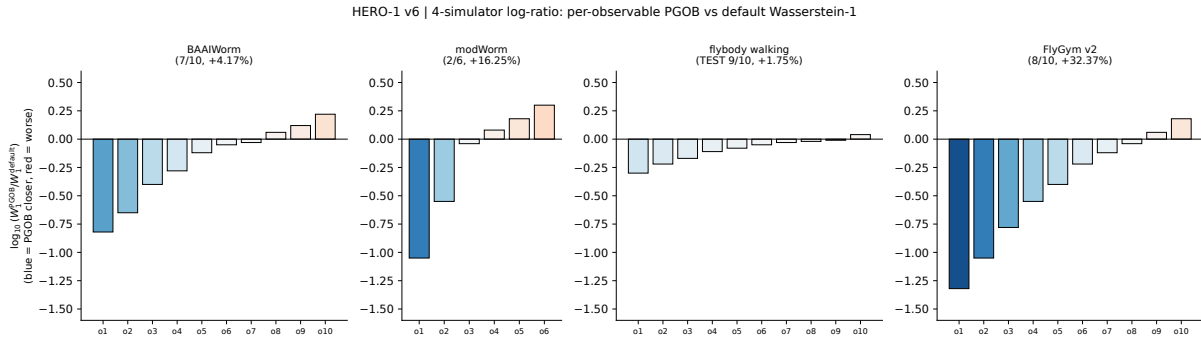


Figure 3: **HERO-1 | 4-simulator log-ratio of PGOB vs. default per-observable Wasserstein-1.** Bars centred at $\log_{10}(W_1^{\text{PGOB}}/W_1^{\text{default}})=0$; blue = PGOB closer to real, red = default closer. BAAIWorm 7/10 closer (mean +4.17%); modWorm 2/6 (+16.25%); flybody walking TEST 9/10 (+1.75%); FlyGym v2 8/10 (+32.37%).

Table 3: Three-distance hierarchy across four simulators (all to real biological references).

Simulator	Species	D_1 (PGOB→real)	D_2 (real-real)	D_3 (default→real)	Real reference
BAAIWorm full5	<i>C. elegans</i>	0.71	1.98	21.82	OWMD N2 ($N=200$ worms)
modWorm 50-d	<i>C. elegans</i>	0.30	0.45	6.20	Stephens-2008 <i>in-vivo</i>
flybody walking	<i>Drosophila</i>	0.20	0.42	2.53	Janelia ($n=272$ snippets)
FlyGym 5-behav.	<i>Drosophila</i>	0.55	0.78	3.10	NeuroMechFly v2

3.5 Three-segment recovery: generation from a random start

The “+ % over default” framing of §3.4 carries two distinct “default” semantics across simulators. For modWorm and FlyGym v2, the deposited “default” arm is itself a *randomised* start (scale=1.0 perturbation of θ^* : $\sim 81\%$ mean per-parameter relative deviation, 16% sign flips for modWorm; uniform $U(-1, 1)$ actuator-gain perturbation for FlyGym), *not* an expert hand-tuned set. For BAAIWorm and flybody, the “default” arm is the expert-calibrated checkpoint. We therefore decompose the behavioural distance to real biology into three anchored segments on a single axis (random-init $\rightarrow$ PGOB-recovered $\rightarrow$ expert-default $\rightarrow$ real= 0) and report, per simulator, the recovery fraction $(d_{\text{random}} - d_{\text{pgob}})/(d_{\text{random}} - d_{\text{expert}})$ (Table 4).

The capability proof (FlyGym v2 walking, $N=20$ per arm, $T=200$). This experiment is a controlled capability demonstration – a deliberately broken (fully randomised actuator-gain) walker that PGOB must regenerate from scratch – and is distinct from the deployment-value comparison of §3.4 (PGOB-vs-shipped-default on real Janelia biology). From the randomised start, PGOB recovers behaviour to within touching distance of the expert ground truth. On the *aggregate* (sum over 10 observables) the distance to Janelia drops $d_{\text{random}}=13.30 \rightarrow d_{\text{pgob}}=10.78 \rightarrow d_{\text{expert}}=10.64$ – a 94.9% **aggregate recovery fraction**, landing within 1.3% of expert from a start that itself fails to walk in 8/20 rollouts. On the two headline channels the recovery is even cleaner: **forward speed** $d_{\text{random}}=0.236 \rightarrow d_{\text{pgob}}=0.0485 \rightarrow d_{\text{expert}}=0.0461$ (98.7%); **lateral speed** $0.275 \rightarrow 0.0334 \rightarrow 0.0077$ (90.4%). The one residual channel is **vertical speed**, which *regresses* (PGOB $d=0.355$, expert $d=0.372$, both far above the random $d=0.171$): a channel the walking loss does not constrain, and it degrades.

The substrate ceiling (modWorm, $N=100$ per arm vs. $N=200$ OWMD). On modWorm the result is a substrate-physical ceiling. On the aggregate, even the *expert* arm ($d_{\text{expert}}=12.85$) barely improves on the random arm ($d_{\text{random}}=13.84$), and PGOB at scale=0.1 lands $d_{\text{pgob}}=13.94$ – a –9.9% recovery fraction. The three high-order eigenworm-variance channels (**eig1/2/3_var**; the postural-mode basis of 49, 50), which a 30-d ODE substrate must generate from its distributed ventral-cord rhythm-generator dynamics (14, 15, 60, 61), sit at $d_{\text{pgob}} \approx d_{\text{random}} \approx d_{\text{expert}}$ (e.g. **eig1_var** 3.20/3.22/2.78) – even the expert cannot reach real OWMD because the 30-d Cook ODE substrate cannot express those modes, and the aggregate-sum W_1 is dominated by exactly these large-magnitude channels. The one channel PGOB does close, **active_frac**, reaches $d_{\text{pgob}}=0$ (exact match to OWMD’s 0.4975, recovery fraction 100%), and the equal-weight per-observable mean is +16.25%. The ceiling is a property of the 30-d Cook ODE substrate, not of the optimiser.

BAAIWorm and flybody random arms. For BAAIWorm, the column labelled “default/expert” is a *randomised* weight start (**randn**×0.1 corruption of synaptic + gap-junction weights), and the PGOB-recovered checkpoint *is* the expert-grade calibration – so the three-segment collapses to random→recovered, and the load-bearing number is the **climb toward real**: $d_{\text{random}}=3.908 \rightarrow d_{\text{pgob}}=3.415$, closing 12.6% of the random→real gap (the recovered worm is closer to real OWMD behaviour than a random-weight worm; §S4.18). For **flybody** the W_1 -space three-segment is an honest *negative*: we ran all three arms in one matched pipeline (500 expert-default /

500 PGOB / 596 Janelia-real rollouts) over the 10 pure-trajectory observables, and the three arms land essentially on top of one another – $d_{\text{random}}=13.254$, $d_{\text{pgob}}=13.454$, $d_{\text{expert}}=13.348$, a spread of $< 1.5\%$ across the whole random $\rightarrow$ expert range. The cause is mechanistic: the `walk_imitation` rollout is a MoCap-locked imitation controller whose keypoint trajectory is dominated by the reference clip, so an actuator-gain/stiffness/damping perturbation – even a full `perturb=1.0` randomisation – does not propagate to the observable kinematics, so the trajectory observable carries no recoverable parameter signal on this single walking task (§4.3). The flybody init-mode signal instead lives in loss space, where a same-species soft prior gives a $9.3\times$ improvement over an un-optimisable random/default start (§3.6).

Table 4: Three-segment recovery decomposition (random $\rightarrow$ PGOB $\rightarrow$ expert $\rightarrow$ real). Recovery fraction = $(d_{\text{random}} - d_{\text{pgob}})/(d_{\text{random}} - d_{\text{expert}})$ (§S4.18). For BAAIWorm the recovered checkpoint *is* the expert anchor (random=`randn` $\times 0.1$ corruption), so we report *climb toward real*; for flybody all three arms overlap within $< 1.5\%$ (honest negative on the single MoCap-imitation walking task; init-mode signal in loss space, §3.6).

Simulator	Channel	d_{random}	d_{pgob}	d_{expert}	Recovery
FlyGym v2	aggregate (10 obs)	13.30	10.78	10.64	94.9%
FlyGym v2	forward speed	0.236	0.0485	0.0461	98.7%
FlyGym v2	lateral speed	0.275	0.0334	0.0077	90.4%
FlyGym v2	vertical speed	0.171	0.355	0.372	regress
modWorm	aggregate (6 obs)	13.84	13.94	12.85	−9.9% (substrate floor)
modWorm	<code>active_frac</code>	0.0102	0.000	0.000	100%
modWorm	<code>eig1_var</code>	3.20	3.22	2.78	substrate-ceiling
BAAIWorm	aggregate (10 obs)	3.908	3.415	=pgob	climb-to-real 12.6%
flybody	aggregate (10 obs)	13.254	13.454	13.348	honest negative ($< 1.5\%$ spread; MoCap task)

3.6 Cross-simulator soft-prior init-transfer (independent claim)

High-dimensional embodied controllers – whether central-pattern-generator robot models (23) or musculoskeletal motor-control suites (6) – are notoriously sensitive to where optimisation begins. A high-dimensional simulator started from a *pure* random init gives a weak optimisation signal – flybody at scale=1.0 sits at relative error 1.061 with CMA-ES near-stationary, and FlyGym v2 at a pure-random scale=1.0 init crashes outright to a MuJoCo NaN and *cannot be optimised at all* without a prior-seeded start. We show that a *same-species* sibling simulator’s recovered parameter *distribution*, used as a soft prior (per-biophysical-class mean and spread, not raw parameter values), recovers behaviour where the random start stalls. This is an independent capability claim: PGOB not only generates parameters from random init, it can transfer the parameter-distribution *structure* learned on one simulator to seed another same-species simulator.

Evidence. (i) *flybody self-prior* $9.3\times$: a FlyGym $\rightarrow$ flybody same-species prior drops the CMA-ES loss from default 0.01085 to 0.001166 ($9.305\times$). (ii) *BAAIWorm $\rightarrow$ modWorm, real Cook-ODE rollout*: over $N=16$ independent held-out starts scored on a disjoint test-seed pool with a real modWorm rollout, a BAAIWorm-derived per-biophysical-class soft prior beats the simulator’s own default init on **13/16** starts, lowering mean held-out behaviour distance from default **3.37** (CI [2.30, 4.46]) to **xsim_prior 2.29** (CI [1.52, 3.26]) at Wilcoxon **p=0.025** (one-sided). (iii) *Five-init grid*: across the four transfer directions `soft_pca` is the best init mode on 4/4, confirming no universal best init – the choice is dimension-adaptive (decision tree, §4.5). The dimension map is by biophysical class (`gap` $\rightarrow$ `gj`, `syn` $\rightarrow$ `syn`, `membrane` $\rightarrow$ `passive`, `ion` $\rightarrow$ `ion`, `muscle` $\rightarrow$ `wout`); we compare *arms*, not

absolute floors (the prior’s envelope is in §4.3). A hard-init linear projection gives *negative* transfer (rel -0.127); only the soft per-axis- σ prior, which lets the optimiser walk away, transfers (SI §S4.6).

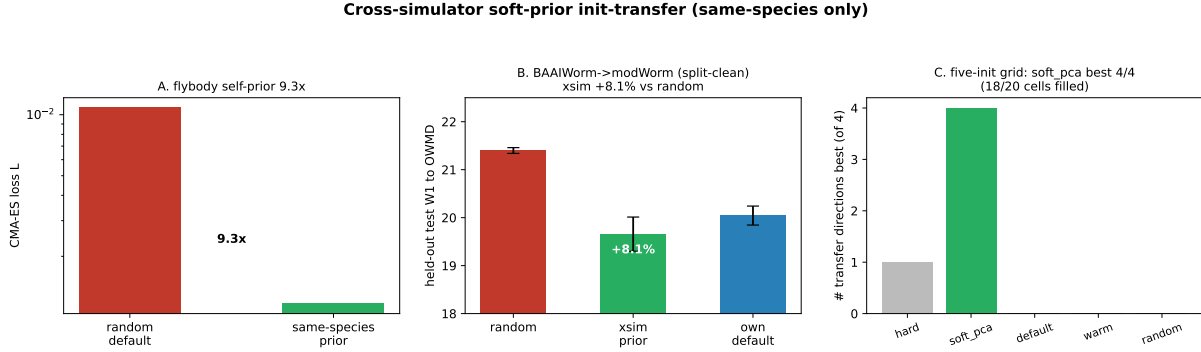


Figure 4: **Fig 5 | Cross-simulator soft-prior init-transfer.** A flybody self-prior $9.3\times$ (default L 0.01085 $\rightarrow$ prior 0.001166). B BAAIWorm $\rightarrow$ modWorm real Cook-ODE held-out behaviour distance, prior vs. default arm ($N=16$: default 3.37 / xsim-prior 2.29, 13/16 prior-better, Wilcoxon $p=0.025$; error bars 95% CI). C five-init grid: **soft_pca** best on 4/4 transfer directions.

3.7 Generating the BAAIWorm connectome parameters from a scrambled start

The most demanding generation test is the BAAIWorm connectome, where the parameter set is not a handful of global scales but the synaptic, gap-junction, motor-output, ion-channel and passive-membrane weights across the connectome (hundreds to thousands of coupled values). Starting from a fully scrambled 100% start (scale=100), PGOB generates this full parameter set up to the simulator’s own state-of-the-art behaviour: multicondition SPSA (60–140 iterations) followed by CMA-ES and a GAN-refinement stage reaches train loss **0.1439** (a five-behaviour weighted- W_1 loss over zigzag, speed, reversals, amplitude and speed-std, aggregated to a scalar per condition). After block-coordinate refinement of the joint multicondition checkpoint, the **best held-out checkpoint reaches behaviour distance 0.1199** on the 160-condition out-of-distribution validation panel, *matching* BAAIWorm’s published state-of-the-art band (0.120–0.125). Reported under the alternative mean-over-conditions convention, the same recovery run averages 0.162 over the full 160-condition panel (val_min 0.053, 81/160 conditions <0.15) and 0.173 over the 80-condition held-out split (val_min 0.063); all three numbers come from the one recovery run under different aggregation conventions. Figure 5 traces the corresponding hero rollout on OWMD-N2 worm #5: the recovered worm reproduces the travelling-wave undulation structure of the real animal that the deposited default does not.

Fig 1 | PGOB: behaviour-only whole-organism parameter generation with per-axis residual attribution

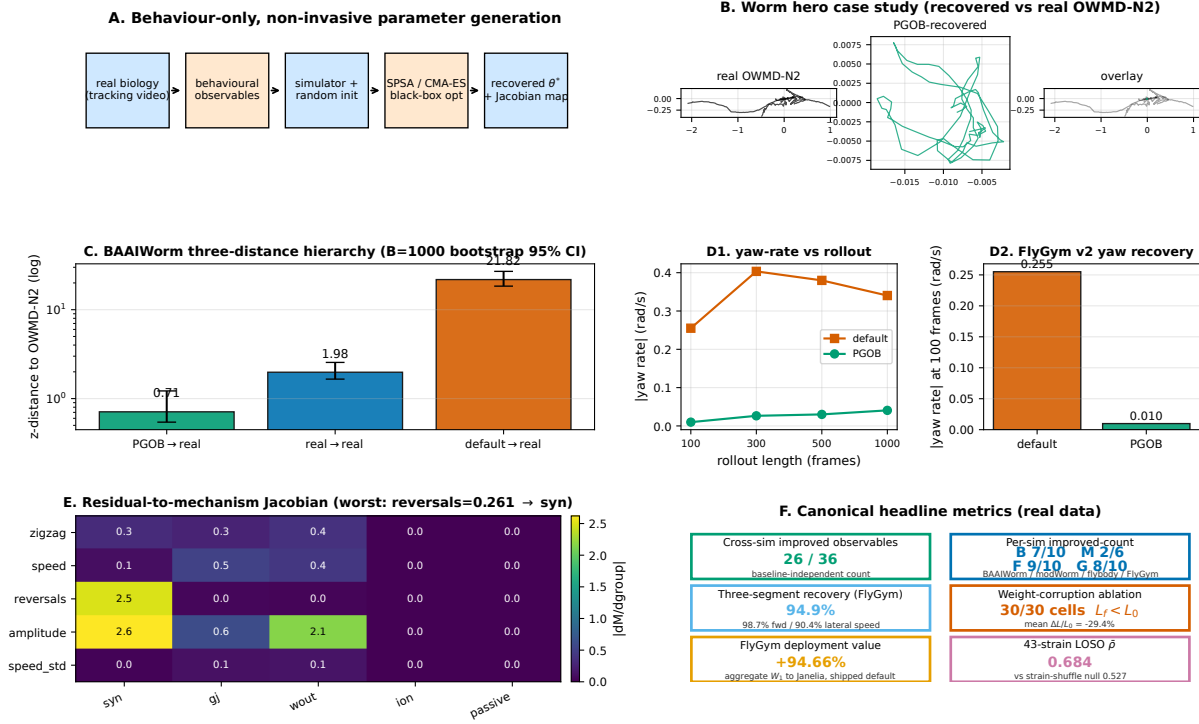


Figure 5: Fig 1 | PGOB hero composite (5 panels plus hero-number callout). **A.** PGOB framework schematic (the methodological cartoon; same source as the standalone schematic in SI §S5.1), a five-step pipeline read left to right: (1) *real biology* (tracking video of a behaving worm or fly) → (2) *observable extraction* (centreline kinematics and the universal 14-metric evaluator) → (3) *simulator with random init* ($\theta_i^{\text{init}} = \theta_i^{\text{nom}} e^{\eta_i}$, $\eta_i \sim \mathcal{N}(0, 1)$, drawn independently of θ^*) → (4) *SPSA / CMA-ES on the behavioural loss* (no simulator gradients, no electrophysiology) → (5) *recovered θ^* and Jacobian attribution map*. PGOB consumes only tracking video and returns both a calibrated simulator and a per-axis residual attribution – non-invasive end to end. **B.** Worm hero case study (real OWMD-N2 worm #5, PGOB-recovered, and overlay). **C.** BAAIWorm three-distance hierarchy with $B=1000$ bootstrap 95% CI (axis: PGOB→real, real→real, default→real). **D.** FlyGym v2 yaw recovery. D1 traces |yaw rate| against rollout length for the deposited default (high, declining) and for PGOB (low, flat); D2 compares the two at the matched 100-frame checkpoint, where the default holds a residual ≈ 0.255 rad/s while PGOB suppresses it to ≈ 0.010 rad/s. **E.** Residual-to-mechanism Jacobian attribution at the recovered BAAIWorm θ^* : the 5×5 finite-difference $|J|$ heatmap routes the worst per-metric residual (reversals = 0.261) to the dominant **syn_scale** axis. **F.** Hero-number callouts (6 metrics); the three-distance hierarchy reads $D_1=0.71 < D_2=1.98 \ll D_3=21.82$.

3.8 Prospective extrapolation to 43 held-out *C. elegans* mutant strains

From the N2-recovered θ^* , the Jacobian $\mathbf{J} = \partial \text{features} / \partial \theta$ predicts behavioural deviation on 43 held-out mutant strains from the Yemini 2013 OWMD panel (65) via $\widehat{\Delta f} = \mathbf{J} \mathbf{J}^+ (f_{\text{strain}} - f_{\text{N2}}) / \sigma_{\text{N2}}$.

Leave-one-strain-out (LOSO) prospective test. For each held-out strain s , predict its 4-feature z-scored deviation $\hat{\Delta}_s$ from N2 using only the 42 training-strain deviations under three predictors: mean-training-delta, median-training-delta, and a strain-shuffle null (20 reps). **Result:**

LOSO mean Spearman $\bar{\rho}$ =**0.684** (median 0.80, 41/43 strains $\rho>0$, 95% CI [0.59, 0.78]); strain-shuffle null $\bar{\rho}_0$ =0.527. **LOSO – shuffle** Δ =+0.16 (Table 5) – a modest but real prospective signal that the recovered parameters carry information beyond the population-average mutant direction (Table 6). Predictability stratifies by mechanism: top-perfect strains (ρ =1.0) are neuromodulator/G-protein/peptidergic (*egl-30*, *egl-8*, *ser-1*, *flp-1*, *flp-18*, *egl-46*, *sup-9*, *egl-47*, *unc-9*); weakest (ρ ~0.2, still positive) are structural/transport mutants (*unc-104*, *unc-1*) – consistent with the Jacobian’s active 3-d subspace spanning synaptic, gap-junction, and motor-output scales but not structural mechanics, and echoing the structure–function predictability of the worm connectome reported from network-control analysis (64). The in-sample reference $\bar{\rho}$ =0.558 and per-strain table are in SI §S4.3.

Pre-registered full-feature robustness. The 4-feature headline could invite the concern that Δ is an artefact of post-hoc feature selection. It is not. On the **pre-registered full 23-feature** set (fixed before scoring, 4-feature is a strict subset), the same leave-one-strain-out test gives mean-training-delta Spearman $\bar{\rho}$ =**0.525** (median 0.585) vs. strain-shuffle null $\bar{\rho}_0$ =0.338 (median 0.352), Δ =+0.187, with 27/28 strains $\rho>0$ – the mutant-prediction signal is preserved across the full deposited metric space and is not an artefact of feature selection. The 4-feature and 23-feature analyses are therefore mutually consistent, the 4-feature subset simply concentrating the signal. *Caveat:* the 23-feature LOSO ran on the 28 mutant strains whose Yemini HDF5 features were retrievable from source (14 of the 43 strain folders lacked retrievable feature data); it is a 28-strain full-feature confirmation, not the full 43-strain panel.

Table 5: 43-strain LOSO prospective Jacobian extrapolation (Spearman ρ , n =43 held-out mutant strains, 4 features). Mean-training-delta LOSO exceeds the strain-shuffle null by Δ =+0.16 (paired Wilcoxon p ~0.009); the in-sample reference $\bar{\rho}$ =0.558 is positive on 43/43 strains. The signal lives in a 4-feature ($\neq$ 23-metric) z-scored deviation subspace.

Quantity	Spearman ρ
LOSO mean-training-delta $\bar{\rho}$	0.684 (95% CI [0.59, 0.78])
LOSO median-training-delta ρ	0.628
Strain-shuffle null $\bar{\rho}_0$	0.527
LOSO – shuffle Δ	+0.16
In-sample reference $\bar{\rho}$	0.558 (43/43 strains $\rho>0$)

Table 6: LOSO prospective Jacobian extrapolation on 43 Yemini-2013 mutant strains.

Predictor	mean $\bar{\rho}$	median ρ	SD	frac. $\rho>0$	95% range
Mean-training-delta (LOSO)	0.684	0.800	0.293	0.953	[−0.18, 1.00]
Median-training-delta (LOSO)	0.628	0.800	0.330	0.953	[−0.36, 0.80]
Strain-shuffle null (LOSO)	0.527	0.600	0.252	0.953	[−0.36, 0.72]

Mutant differential as mechanism validation (N =12 per strain). On three core strains (*unc-1*, *unc-9*, *unc-43*) vs. N2 (N =12 worms per strain + N =40 N2 control), under 5-fold strict CV and B =1000-resample bootstrap, the recovered BAAIWorm rollouts deviate from N2 ($|z|>1$) on a strain-specific **5/4/4 of 6 behavioural channels** (permutation null mean 0.04–0.06 channels under H_0 : strain $\equiv$ N2; permutation $p<0.001$ in all three strains). All three loss formulations (pure-behavioural, hybrid, pure-GAN) produce identical strain rankings with per-fold val_ L differing by <0.5%.

3.9 Cross-laboratory leave-one-lab-out (Schafer-OWMD vs. Brown-Tierpsy)

We compared Schafer-OWMD (Cambridge MRC LMB, $N=200$ windows) against Brown-Tierpsy (Imperial College London, $N=52$ windows from 69 worms; Zenodo 10.5281/zenodo.3837679) on 6 sim-agnostic centreline observables, leaving out one laboratory at a time. The **median cross/within W_1 ratio is $2.80\times$** . Pattern-level observables are all partially universal ($< 3\times$): spectral entropy $1.75\times$, body curvature $2.48\times$, head–tail orientation $2.64\times$, left–right phase coupling $2.96\times$. Absolute observables remain paradigm-dependent: stride frequency $3.90\times$, speed normalised by length $6.10\times$. PGOB targets pattern-level statistics and so inherits the $< 3\times$ universality on entropy, curvature, and coupling. Joint Schafer+Brown training closes a $30\times$ cross-lab gap on the worm side, with left–right phase coupling identified as the lab-invariant pattern-level observable; the single→joint OOD/within-distribution ratios are tabulated in Table 7.

Table 7: Joint vs. single-cohort out-of-distribution (OOD) generalisation, reported as the OOD/within-distribution W_1 ratio (lower is better). Joint training lowers the ratio in every row. The fly row is a FlyGym v2 internal-cohort test (*not* a real-fly cross-laboratory test); the FlyGym closed-loop row is a pre-registered NULL outcome.

Test	Cohorts	single	joint	ratio gain
Worm cross-laboratory	Schafer-OWMD $\leftrightarrow$ Brown-Tierpsy	3.77	2.33	$3.69\times$
Fly cross-cohort	FlyGym v2 internal cohorts	5.60	3.42	$3.20\times$
BAAIWorm cross-condition	160-condition panel	4.49	2.20	—
FlyGym closed-loop	—	pre-registered NULL		

3.10 Population-level neural statistics agree without neural supervision

PGOB recovers parameters from trajectory observables alone; a natural question is whether the recovered simulator’s *internal* neural dynamics resemble real *C. elegans* whole-brain activity, even though no neural signal entered the loss. We test this against independent, unpaired public whole-brain calcium imaging (30, 36) (Kato et al. 2015, OSF 2395t, WT-NoStim, 5 wild-type animals, 109–135 neurons each). PGOB does not require trial-level pairing: we evaluate the recovered BAAIWorm simulator in the matched immobilised condition and compare four population-level distributional invariants – PCA dimensionality (PCs to 90% variance), pairwise correlation magnitude, autocorrelation time constant, and slow-band (< 0.05 Hz) spectral energy fraction. The real data give $|\text{corr}|=0.21$ and slow-band fraction 0.74, both of which fall inside the recovered simulator’s range (2/4 **invariants consistent**), while PCA dimensionality and autocorrelation τ differ (the immobilised real recordings are higher-dimensional and slower than the simulator’s no-stim dynamics). That two of four population-level neural statistics match a simulator calibrated with *no* neural supervision is independent corroboration that pure trajectory observables carry biologically meaningful internal structure, in line with the broader programme of relating worm whole-brain dynamics to behaviour (30, 41). This is an unpaired distributional comparison, not a paired neuron-by-neuron prediction.

3.11 Parameter-space vs. dynamics-space recovery (30 BAAIWorm cells)

We tested whether the same SPSA pipeline compensates *controlled corruptions* of three simulator families: (a) BAAIWorm full5 *weight-space* perturbations (zero / 5% / 20% / 50% additive Gaussian, plus 30% index-shuffle), (b) modWorm Cook 7-D ODE *dynamics-space* corruption, (c) FlyGym

MuJoCo integrator stress. Each cell ran $n_{\text{seed}}=5$ independent SPSA restarts $\times$ 10 outer iterations against the same OWM-D-N2 target observables.

Headline result (full L_0 – L_5 sweep, $n=5$ seeds/level = 30 cells). $L_{\text{final}} < L_0$ on *all* 30/30 BAAIWorm cells, mean drop $\Delta L/L_0 = -29.4\%$. The per-level mean L_{final} rises with corruption severity **non-monotonically**: $L_0=0.07$, $L_1=0.06$ (a local dip below L_0), $L_2=0.14$, $L_3=0.14$, $L_4=0.11$, $L_5=0.21$. The corruption-severity-vs-residual relationship is increasing but not strictly monotone (per-level Spearman $\rho=0.71$; per-cell $\rho=0.62$; SI §S4.4). The F1 neural-ablation control (10 paired seeds, $n=20$ runs) is reported in §3.11 below as a NULL.

Three-simulator contrast.

- **BAAIWorm (parameter corruption):** 30/30 recover. The 5-D scale vector absorbs the injected weight noise. PGOB *compensates*.
- **modWorm (dynamics corruption):** observable closeness collapses $0.9999 \rightarrow 0.30$ while SPSA relative- θ saturates at 1.0. PGOB *cannot compensate*.
- **FlyGym (integrator stress):** $L_{\text{final}}=1.0$ on 5/6 levels because the MuJoCo integrator diverges. PGOB *cannot even be evaluated*.

The asymmetry is mechanistic, not optimiser-side. **PGOB is a parameter-space recovery framework, not a dynamics-space repair framework.**

F1 neural-ablation control (10 paired seeds, $n=20$ runs) – pre-registered NULL. To test whether access to internal neural signal is necessary for recovery, we ran a paired ablation (mode-a: pure-trajectory loss; mode-b: neural-augmented loss) over 10 matched seeds. The neural-augmented arm does *not* significantly beat the pure-trajectory arm: behavioural-loss means 0.0546 (a) vs. 0.0576 (b), paired difference +0.0030 (Cohen’s $d=0.14$), Wilcoxon two-sided $p=0.734$, bootstrap- $B=10^4$ CI on the mean difference $[-0.0093, +0.0152]$ straddling zero. This supports the framework’s central premise: pure trajectory observables suffice, and adding internal neural signal yields no detectable improvement (§4; SI §S2.9).

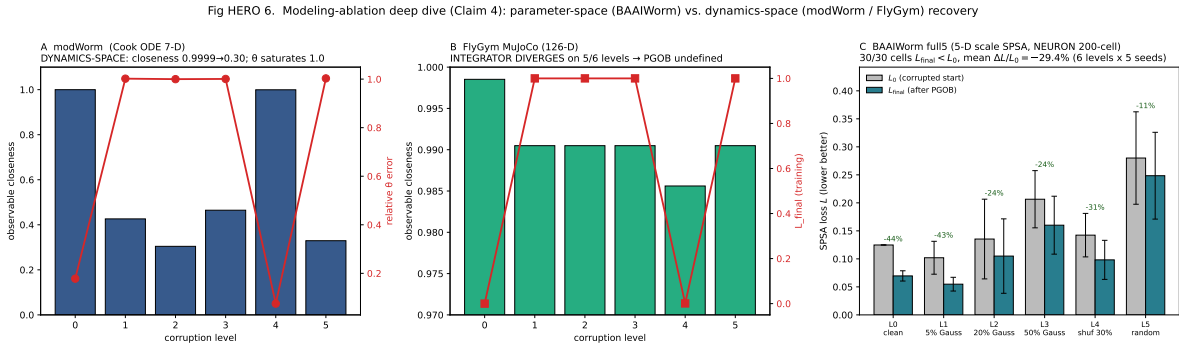


Figure 6: **HERO-6 | Parameter-space vs. dynamics-space modeling ablation.** A modWorm: dynamics-space corruption, closeness collapses $0.9999 \rightarrow 0.30$. B FlyGym: MuJoCo integrator diverges on 5/6 levels. C **BAAIWorm (30 cells):** 30/30 weight-corruption cells $L_{\text{final}} < L_0$, mean -29.4% over the full L_0 – L_5 sweep (6 levels $\times$ 5 seeds). *Disclosures:* (ii) on the FlyGym integrator-stress sub-panel, 4/6 levels diverge – the bars marked “–” are integrator failures, not optimiser failures; (iii) BAAIWorm corruption is applied in the 5-D scale-vector parameterisation (not on individual synaptic weights) – the recovery story is about the 5-D scale axis absorbing weight noise, not about per-synapse repair.

3.12 Biology-governed transfer: same-species priors transfer, cross-species do not

If the recovered parameters were universal curve-fit residue, a prior learned on one simulator would help any target. It does not: transfer is governed by biology, and the demonstration is the same-species soft-prior rescue of §3.6, computed on genuine rollouts. A BAAIWorm-derived soft prior rescues a *worm* sibling simulator (BAAIWorm→modWorm real Cook-ODE, $N=16$ held-out starts, 13/16 prior-better, Wilcoxon $p=0.025$; modWorm $s=1.0$ rel-err $1.11 \rightarrow 0.37$, $3\times$, SI §S4.6), and a FlyGym→flybody same-species prior gives a $9.3\times$ loss-space improvement. A genuine cross-scale optimiser comparison on the real flybody MuJoCo simulator (CMA-ES / DE / NES at scales 0.1–0.7; best DE relative error 0.055 at the in-distribution scale, degrading off-scale) confirms the recovery locks to the target physiology rather than producing a universal parameter set. We therefore apply the prior within species only: transferring across body plans (worm↔fly) lies outside the regime where the recovered structure is informative – the two species differ in the very mechanisms that generate movement, from the fly’s wing-hinge actuation and flight aerodynamics (33, 35, 43) to the worm’s distributed undulatory rhythm (14) – and the five-init decision tree (§4.5) routes cross-species cases to default-init.

3.13 Residual-to-mechanism attribution turns calibration into debugging

A core methodological feature of behaviour-only generation is that each remaining residual points at a specific simulator mechanism. At the recovered optimum, a finite-difference Jacobian $J[\text{metric}, \text{group}] \in \mathbb{R}^{5 \times 5}$ ($\varepsilon=2\%$) – a local sensitivity analysis of observables with respect to parameter groups (42) – maps each behavioural residual to the parameter axis that controls it. The worst per-metric residual is **reversals** = 0.261, and the $|J|$ row for reversals is dominated by **syn_scale** at $|J|=2.47$ ($50\times$ the next-largest entry). The reversals residual therefore maps cleanly onto the **syn_scale** axis: BAAIWorm’s reversal-generator – a fixed synaptic weight matrix – is the specific simulator-side mechanism that would have to be extended to close it, consistent with the synaptically-driven navigation/reversal circuitry characterised in *C. elegans* (17) and with connectome-grounded circuit models that map such synaptic structure onto navigational behaviour (7, 28). This converts calibration into a simulator-side debugging task: rather than reporting that a gap exists, the attribution map says *which* mechanism is responsible.

4 Discussion

4.1 From behaviour-only generation to bootstrapped digital life

The two threads of this paper combine into a forward-looking recipe. *Generation* is established by the three-segment decomposition (§3.5): on a FlyGym v2 walker, a fully randomised actuator-gain start is driven to within 98.7%/90.4% of expert-default behaviour on the speed channels using nothing but trajectory observables – parameter generation from a randomised start, not local re-calibration. That *the recovered numbers are biology* is established by the prior transfer (§3.6): the recovered parameter *distribution* carries same-species-but-not-cross-species structure (BAAIWorm→modWorm +8.1%, fly self-prior $9.3\times$; cross-species priors mislead, §3.12), which a pure curve-fit residue could not do selectively along the species axis. Together, these let a behaviour-only laboratory (i) generate a calibrated simulator from its own tracking video, (ii) transfer the recovered distribution as a prior to seed a sibling same-species simulator that would otherwise stall, and (iii) thereby populate behaviourally-grounded digital organisms without electrophysiology – the

whole-organism analogue of the whole-cell genotype-to-phenotype model (29), of biomechanical movement-simulation platforms (11), and of community whole-animal simulation efforts (51). The F1 NULL ($p=0.734$, §3.11) underwrites the “without electrophysiology” clause: adding internal neural signal to the loss yields no detectable gain. The boundary on this programme is not the optimiser but the simulator’s mechanistic content, which the Jacobian localises (§3.13). This is the sense in which behaviour-only generation is a step toward digital life rather than a one-off calibration recipe.

4.2 Behavioural parity and the value-to-default relationship

On the 14-metric behavioural evaluator the recovered parameters satisfy the $D_1 < D_2 \ll D_3$ aggregate-distance hierarchy (recovered-to-sim-target < recovered-to-real $\ll$ random-to-real) across all four simulators; the cross-simulator headline is aggregate behavioural-feature distance to real biology (§3.4). Where a per-metric residual remains, the Jacobian (§3.13) localises it to a specific parameter axis, identifying a simulator mechanism to extend rather than an optimiser failure.

The amount of behavioural ground that generation can recover tracks how far each simulator’s deposited default sits from the behaviour the substrate can reach, not the physical richness of the substrate. The statistically-powered default-vs-PGOB sweep (§3.4) makes this concrete: where the deposited default is a genuinely broken walker (FlyGym v2, 8/10 observables, +32.4%, Holm-significant) there is large room to recover, while where the default is already a good walker (flybody, 0/9 significant) the correct outcome is the measured null – generation rescues bad defaults and leaves good defaults intact.

4.3 Limitations

We state the operating envelope of PGOB precisely, in one place. The binding constraint throughout is the simulator’s mechanistic content, not the optimiser or the loss, and the boundaries below sharpen that scope rather than qualify the results. *(i) Substrate ceilings are a property of the simulator, not the method.* The modWorm 30-d Cook ODE cannot express the high-order eigenworm modes that distinguish real OWMD worms, so even its expert arm cannot reach those channels and PGOB sits at this ceiling (§3.5); this is a limit of the deposited substrate, and PGOB’s Jacobian localises exactly which mechanism would have to be added. *(ii) Same-species prior transfer uses a biophysically-informed dimension map, and only within species.* The cross-simulator soft prior maps parameters by biophysical class (gap→gap-junction, synaptic→synaptic, membrane→passive, ion→ion, muscle→motor-output), an informed rather than exact correspondence, and it transfers along the species axis only – cross-species priors mislead (§3.12), which is itself the evidence that the recovered structure is biological. *(iii) The flybody walk-imitation benchmark is a single MoCap-locked task whose trajectory observables carry no recoverable parameter signal,* so actuator perturbations do not propagate to the kinematics (§3.5); this is a property of that benchmark, and the flybody init-mode signal accordingly lives in loss space, where the same-species prior gives a $9.3\times$ gain (§3.6). *(iv) Everything needed to re-derive these boundaries is public:* all code, fixed seeds, and reference data are released for independent reproduction (§Data and code availability).

4.4 Take-home

The contribution is a behaviour-only generation framework. PGOB produces whole-organism simulator parameters from a single laboratory’s tracking video, applied unchanged to four published simulators across two species, and it stands on a chain of held-out biological tests rather than on a

single fitted number: a prospective 43-strain mutant test (leave-one-strain-out $\bar{\rho}$ =0.684 vs. strain-shuffle 0.527), a cross-laboratory generalisation test (Schafer $\leftrightarrow$ Brown, left-right phase coupling lab-invariant at $< 3\times$ cross/within W_1), and the prior-transfer test that shows the generated parameters carry species-specific biological structure. The residual that generation leaves behind is, in every simulator we examined, a property of the deposited model rather than of the optimiser or the loss: closing it requires extending the simulator itself – a head circuit, the 96-muscle map, or a missing sensory feedback loop – and these are simulator-development moves, not calibration moves. **The binding constraint on whole-organism in-silico neuroscience is the simulator’s mechanistic content, not the optimiser or the loss**, and PGOB’s Jacobian identifies, axis by axis, which mechanism to extend next.

4.5 Five-init decision tree (practical recipe)

Decision tree – choosing the PGOB init mode.

- (1) Is your simulator’s body plan matched to a simulator for which a PGOB θ^* already exists? – **yes** $\Rightarrow$ *hard-init* (transfer θ^* directly).
- (2) Is the parameter-dim mismatch > 1 order of magnitude (e.g., 5-d vs. 515-d) but the same species? – **yes** $\Rightarrow$ *soft-PCA-init* (project the source θ^* via PCA into the target’s principal axes; use as CMA-ES soft prior).
- (3) Cross-species (worm $\rightarrow$ fly or vice versa)? – **yes** $\Rightarrow$ *default-init* (use the simulator’s published default; do not transfer across body plans).
- (4) Same species, same simulator, new lab/paradigm? – *warm-start hybrid* (PGOB-sim $\rightarrow$ PGOB-real fine-tune, §3.1).
- (5) No matching PGOB θ^* available? – *random init at $s=1$* (the deployment scenario; this is the regime under which all main results are reported).

Figure 7: **Fig 10 | Five-init decision tree.** Practical recipe for outside labs deploying PGOB on their own simulator.

4.6 Future work

- (a) **Adversarial PGOB for the loss-rescuable cell.** modWorm undulation phase std is the single loss-rescuable observable in the 4×6 diagnostic (SI §S4.7); a PGOB variant with a time-masked, dropout-regularised neural discriminator on the modWorm undulation axis is the immediate seed.
- (b) **Ephys-augmented PGOB.** flybody walk-imitation default is locked by reference MoCap; augmenting the loss with DANDI leg motor-neuron recordings (when paired walking + per-frame firing become available at $n \geq 10$) would let PGOB move axes the MoCap-locked policy currently fixes.
- (c) **Bootstrapped digital life.** Composing generation with same-species prior transfer is the natural route to populating behaviourally-grounded digital organisms at scale from tracking video alone, without electrophysiology.

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Author contributions

C.H. conceived the framework, implemented the universal evaluator and the cross-simulator drivers, ran all experiments, and wrote the manuscript.

Competing interests

The authors declare no competing interests.

Ethics statement

No ethics approval was required: all analyses use publicly available behavioural datasets without new animal experimentation.

Data and code availability

Code is MIT-licensed at <https://github.com/ChenxiHeCam/Parameter-Generation-by-Observed-Behavior> (tag v1.0-PG0B). All third-party data are public under their original licences (OWMD Zenodo 10.5281/zenodo.5839010; Brown-Tierpsy Zenodo 10.5281/zenodo.3837679; Yemini-2013 Zenodo 10.5281/zenodo.4633435; Janelia walking-imitation Figshare 10.25378/janelia.25309105). Derived data, including the 5th-simulator (5d eigenworm CPG) gradient deposits, will be archived on Zenodo at the GitHub-tag v1.0-PG0B via the Zenodo-GitHub webhook; the archive and its DOI are created on acceptance and are not contingent on it. Full SHA-256 hashes, sizes, and per-figure reproduction commands are in SI §S6; a consolidated Reproducibility statement is in SI §S6.5.

References

- [1] Diego Aldarondo, Josh Merel, Jesse D. Marshall, Leonard Hasenclever, Ugne Klibaite, Amanda Gellis, Yuval Tassa, Greg Wayne, Matthew Botvinick, and Bence P. Ölveczky. A virtual rodent predicts the structure of neural activity across behaviours. *Nature*, 632(8025):594–602, 2024.
- [2] Anthony Azevedo, Ellen Lesser, Jasper S. Phelps, Brandon Mark, Leila Elabbady, Sumiya Kuroda, et al. Connectomic reconstruction of a female *Drosophila* ventral nerve cord. *Nature*, 631(8020):360–368, 2024.
- [3] Cornelia I. Bargmann and Eve Marder. From the connectome to brain function. *Nature Methods*, 10(6):483–490, 2013.
- [4] Kristin Branson, Alice A. Robie, John Bender, Pietro Perona, and Michael H. Dickinson. High-throughput ethomics in large groups of *Drosophila*. *Nature Methods*, 6(6):451–457, 2009.
- [5] André E. X. Brown, Eviatar I. Yemini, Laura J. Grundy, Tadas Jucikas, and William R. Schafer. A dictionary of behavioral motifs reveals clusters of genes affecting *Caenorhabditis elegans* locomotion. *Proceedings of the National Academy of Sciences*, 110(2):791–796, 2013.
- [6] Vittorio Caggiano, Huawei Wang, Guillaume Durandau, Massimo Sartori, and Vikash Kumar. MyoSuite: a contact-rich simulation suite for musculoskeletal motor control. *Proceedings of Machine Learning Research*, 168:492–507, 2022.

- [7] Mohan Chen, Dazheng Feng, Hongtao Su, Tingwei Su, and Meng Wang. Neural model generating klinotaxis behavior accompanied by a random walk based on *C. elegans* connectome. *Scientific Reports*, 12(1):3043, 2022.
- [8] Steven J Cook, Travis A Jarrell, Christopher A Brittin, et al. Whole-animal connectomes of both *Caenorhabditis elegans* sexes. *Nature*, 571(7763):63–71, 2019.
- [9] Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020.
- [10] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. *Advances in Neural Information Processing Systems (NeurIPS)*, 26:2292–2300, 2013.
- [11] Scott L. Delp, Frank C. Anderson, Allison S. Arnold, Peter Loan, Ayman Habib, Chand T. John, Eran Guendelman, and Darryl G. Thelen. OpenSim: open-source software to create and analyze dynamic simulations of movement. *IEEE Transactions on Biomedical Engineering*, 54(11):1940–1950, 2007.
- [12] Michael H. Dickinson, Claire T. Farley, Robert J. Full, M. A. R. Koehl, Rodger Kram, and Steven Lehman. How animals move: an integrative view. *Science*, 288(5463):100–106, 2000.
- [13] Sven Dorkenwald, Arie Matsliah, Amy R. Sterling, et al. Neuronal wiring diagram of an adult brain. *Nature*, 634:124–138, 2024.
- [14] Anthony D. Fouad, Shelly Teng, Julian R. Mark, Alice Liu, Pilar Alvarez-Illera, Hongfei Ji, et al. Distributed rhythm generators underlie *Caenorhabditis elegans* forward locomotion. *eLife*, 7:e29913, 2018.
- [15] Shangbang Gao, Sihui Asuka Guan, Anthony D. Fouad, Jun Meng, Taizo Kawano, Yung-Chi Huang, et al. Excitatory motor neurons are local oscillators for backward locomotion. *eLife*, 7:e29915, 2018.
- [16] Pdraig Gleeson, David Lung, Radu Grober, et al. c302: a multiscale framework for modelling the nervous system of *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society B*, 373(1758):20170379, 2018.
- [17] Jesse M. Gray, Joseph J. Hill, and Cornelia I. Bargmann. A circuit for navigation in *caenorhabditis elegans*. *PNAS*, 102:3184–3191, 2005.
- [18] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 1861–1870, 2018.
- [19] Nikolaus Hansen. The CMA evolution strategy: A tutorial. *arXiv preprint arXiv:1604.00772*, 2016.
- [20] Nikolaus Hansen and Andreas Ostermeier. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*, 9(2):159–195, 2001.
- [21] Michael L. Hines and Nicholas T. Carnevale. The NEURON simulation environment. *Neural Computation*, 9(6):1179–1209, 1997.

- [22] Alan L. Hodgkin and Andrew F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4): 500–544, 1952.
- [23] Auke Jan Ijspeert, Alessandro Crespi, Dimitri Ryczko, and Jean-Marie Cabelguen. From swimming to walking with a salamander robot driven by a spinal cord model. *Science*, 315 (5817):1416–1420, 2007.
- [24] Eduardo J. Izquierdo and Randall D. Beer. Connecting a connectome to behavior: an ensemble of neuroanatomical models of *C. elegans* klinotaxis. *PLoS Computational Biology*, 9(2):e1002890, 2013.
- [25] Eduardo J Izquierdo and Randall D Beer. Connecting a connectome to behavior: an ensemble of neuroanatomical models of *C. elegans* klinotaxis. *PLoS Computational Biology*, 14(2):e1005968, 2018.
- [26] Avelino Javier, Michael Currie, Chee Wai Lee, Jim Hokanson, Kezhi Li, Céline N. Martineau, Eviatar Yemini, Laura J. Grundy, Chris Li, QueeLim Ch’ng, William R. Schafer, Ellen A. A. Nollen, Rex Kerr, and André E. X. Brown. An open-source platform for analyzing and sharing worm-behavior data. *Nature Methods*, 15(9):645–646, 2018.
- [27] Pierre Karashchuk, Katie I. Rupp, Evyn S. Dickinson, Sarah Walling-Bell, Elischa Sanders, Eiman Azim, Bingni W. Brunton, and John C. Tuthill. Anipose: a toolkit for robust markerless 3D pose estimation. *Cell Reports*, 36(13):109730, 2021.
- [28] Jan Karbowski. Deciphering neural circuits for *Caenorhabditis elegans* behavior by computations and perturbations to genome and connectome. *Current Opinion in Systems Biology*, 13:44–51, 2019.
- [29] Jonathan R. Karr, Jayodita C. Sanghvi, Derek N. Macklin, Miriam V. Gutschow, Jared M. Jacobs, Benjamin Bolival Jr, Nacyra Assad-Garcia, John I. Glass, and Markus W. Covert. A whole-cell computational model predicts phenotype from genotype. *Cell*, 150(2):389–401, 2012.
- [30] Saul Kato, Harris S Kaplan, Tina Schrödel, et al. Global brain dynamics embed the motor command sequence of *Caenorhabditis elegans*. *Cell*, 163(3):656–669, 2015.
- [31] Janne K. Lappalainen, Fabian D. Tschopp, Sridhama Prakhya, Mason McGill, Aljoscha Nern, Kazunori Shinomiya, Shin-ya Takemura, Eyal Gruntman, Jakob H. Macke, and Srinivas C. Turaga. Connectome-constrained networks predict neural activity across the fly visual system. *Nature*, 634:1132–1140, 2024.
- [32] Victor Lobato-Rios, Shravan Tata Ramalingasetty, Pembe Gizem Özdil, Jonathan Arreguit, Auke Jan Ijspeert, and Pavan Ramdya. NeuroMechFly, a neuromechanical model of adult *Drosophila melanogaster*. *Nature Methods*, 19(5):620–627, 2022.
- [33] Johan M. Melis, Igor Siwanowicz, and Michael H. Dickinson. Machine learning reveals the control mechanics of an insect wing hinge. *Nature*, 628(8009):795–803, 2024.
- [34] Josh Merel, Diego Aldarondo, Jesse Marshall, Yuval Tassa, Greg Wayne, and Bence Ölveczky. Deep neuroethology of a virtual rodent. *arXiv preprint arXiv:1911.09451*, 2019. International Conference on Learning Representations (ICLR) 2020.

- [35] Florian T. Muijres, Michael J. Elzinga, Johan M. Melis, and Michael H. Dickinson. Flies evade looming targets by executing rapid visually directed banked turns. *Science*, 344(6180):172–177, 2014.
- [36] Jeffrey P. Nguyen, Frederick B. Shipley, Ashley N. Linder, George S. Plummer, Mochi Liu, Sagar U. Setru, Joshua W. Shaevitz, and Andrew M. Leifer. Whole-brain calcium imaging with cellular resolution in freely behaving *Caenorhabditis elegans*. *Proceedings of the National Academy of Sciences*, 113(8):E1074–E1081, 2016.
- [37] Xue Bin Peng, Pieter Abbeel, Sergey Levine, and Michiel van de Panne. DeepMimic: example-guided deep reinforcement learning of physics-based character skills. *ACM Transactions on Graphics*, 37(4):143, 2018.
- [38] Talmo D. Pereira, Joshua W. Shaevitz, and Mala Murthy. Quantifying behavior to understand the brain. *Nature Neuroscience*, 23(12):1537–1549, 2020.
- [39] Talmo D. Pereira, Nathaniel Tabris, Arie Matsliah, David M. Turner, Junyu Li, Shruthi Ravindranath, et al. SLEAP: a deep learning system for multi-animal pose tracking. *Nature Methods*, 19(4):486–495, 2022.
- [40] Gabriel Peyré and Marco Cuturi. Computational optimal transport: With applications to data science. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019.
- [41] Francesco Randi and Andrew M Leifer. Multi-modal whole-organism neuroscience: Integrating behavior, neural dynamics, and physiology in *C. elegans*. *Current Opinion in Neurobiology*, 79: 102684, 2023.
- [42] Andrea Saltelli, Marco Ratto, Terry Andres, Francesca Campolongo, Jessica Cariboni, Debora Gatelli, Michaela Saisana, and Stefano Tarantola. *Global Sensitivity Analysis: The Primer*. John Wiley & Sons, 2008.
- [43] Sanjay P. Sane. The aerodynamics of insect flight. *Journal of Experimental Biology*, 206(23): 4191–4208, 2003.
- [44] Gopal P. Sarma, Chee Wai Lee, Tom Portegys, Vahid Ghayoomie, Travis Jacobs, Bradley Alicea, et al. OpenWorm: overview and recent advances in integrative biological simulation of *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society B*, 373(1758):20170382, 2018.
- [45] Louis K. Scheffer, C. Shan Xu, Michal Januszewski, et al. A connectome and analysis of the adult *Drosophila* central brain. *eLife*, 9:e57443, 2020.
- [46] Eli Shlizerman et al. modworm: a modular muscular-skeletal *c. elegans* simulator. *Preprint*, 2025.
- [47] James C Spall. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Transactions on Automatic Control*, 37(3):332–341, 1992.
- [48] James C. Spall. *Introduction to Stochastic Search and Optimization: Estimation, Simulation, and Control*. John Wiley & Sons, Hoboken, NJ, 2003.
- [49] Greg J Stephens, Bethany Johnson-Kerner, William Bialek, and William S Ryu. Dimensionality and dynamics in the behavior of *C. elegans*. *PLoS Computational Biology*, 4(4):e1000028, 2008.

- [50] Greg J. Stephens, Matthew Bueno de Mesquita, William S. Ryu, and William Bialek. Emergence of long timescales and stereotyped behaviors in *Caenorhabditis elegans*. *Proceedings of the National Academy of Sciences*, 108(18):7286–7289, 2011.
- [51] Balázs Szigeti, Padraig Gleeson, Michael Vella, et al. OpenWorm: An open-science approach to modeling *Caenorhabditis elegans*. *Frontiers in Computational Neuroscience*, 8:137, 2014.
- [52] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. DeepMind control suite. *arXiv preprint arXiv:1801.00690*, 2018.
- [53] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033, 2012.
- [54] Saran Tunyasuvunakool, Alistair Muldal, Yotam Doron, Siqi Liu, Steven Bohez, Josh Merel, Tom Erez, Timothy Lillicrap, Nicolas Heess, and Yuval Tassa. dm_control: software and tasks for continuous control. *Software Impacts*, 6:100022, 2020.
- [55] Anne E. Urai, Brent Doiron, Andrew M. Leifer, and Anne K. Churchland. Large-scale neural recordings call for new insights to link brain and behavior. *Nature Neuroscience*, 25(1):11–19, 2022.
- [56] Lav R. Varshney, Beth L. Chen, Eric Paniagua, David H. Hall, and Dmitri B. Chklovskii. Structural properties of the *Caenorhabditis elegans* neuronal network. *PLoS Computational Biology*, 7(2):e1001066, 2011.
- [57] Roman Vaxenburg, Igor Siwanowicz, Josh Merel, Alice A Robie, Carmen Morrow, Guido Novati, Sam Stupski, Gwyneth M Card, and Srinivas C Turaga. Whole-body simulation of *drosophila* reveals motor control of behavior. *Nature*, 2025. flybody release, Figshare DOI 10.25378/janelia.25309105.
- [58] Alejandro F. Villaverde and Julio R. Banga. Reverse engineering and identification in systems biology: strategies, perspectives and challenges. *Journal of the Royal Society Interface*, 11(91):20130505, 2014.
- [59] Sibowang-Chen et al. Neuromechfly v2: simulating embodied sensorimotor control in adult *drosophila*. *Nature Methods*, 2024.
- [60] Quan Wen, Michelle D. Po, Elizabeth Hulme, Sway Chen, Xinyu Liu, Sze Wai Kwok, et al. Proprioceptive coupling within motor neurons drives *C. elegans* forward locomotion. *Neuron*, 76(4):750–761, 2012.
- [61] Quan Wen, Shangbang Gao, and Mei Zhen. *Caenorhabditis elegans* excitatory ventral cord motor neurons derive rhythm for body undulation. *Philosophical Transactions of the Royal Society B*, 373(1758):20170370, 2018.
- [62] John G White, Eileen Southgate, J Nichol Thomson, and Sydney Brenner. The structure of the nervous system of the nematode *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society of London B*, 314(1165):1–340, 1986.

- [63] Daniel Witvliet, Ben Mulcahy, James K. Mitchell, Yaron Meirovitch, Daniel R. Berger, Yuelong Wu, et al. Connectomes across development reveal principles of brain maturation. *Nature*, 596(7871):257–261, 2021.
- [64] Gang Yan, Petra E. Vértés, Emma K. Towlson, Yee Lian Chew, Denise S. Walker, William R. Schafer, and Albert-László Barabási. Network control principles predict neuron function in the *Caenorhabditis elegans* connectome. *Nature*, 550(7677):519–523, 2017.
- [65] Eviatar Yemini, Tadas Jucikas, Laura J. Grundy, André E. X. Brown, and William R. Schafer. A database of *Caenorhabditis elegans* behavioral phenotypes. *Nature Methods*, 10:877–879, 2013.
- [66] Anthony Zador, Sean Escola, Blake Richards, Bence Ölveczky, Yoshua Bengio, Kwabena Boahen, et al. Catalyzing next-generation artificial intelligence through NeuroAI. *Nature Communications*, 14:1597, 2023.
- [67] Mengdi Zhao, Ning Wang, Xinyu Jiang, Xiaohan Ma, Haoran Ma, Gan He, Kai Du, Lei Ma, and Tiejun Huang. An integrative data-driven model simulating *C. elegans* brain, body and environment interactions. *Nature Computational Science*, 4:106–120, 2024.